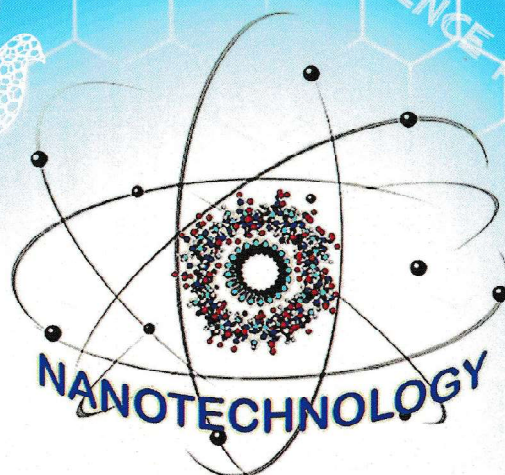




ABSTRACT BOOK

**International research
and practice conference:**

**NANOTECHNOLOGY
AND NANOMATERIALS
(NANO-2026)**

26-28 of August 2026
Chernivtsi, UKRAINE

**INTERNATIONAL RESEARCH
AND PRACTICE CONFERENCE
“NANOTECHNOLOGY
AND NANOMATERIALS”**

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Abstract book

The International research and practice conference “Nanotechnology and nanomaterials” (NANO-2026). Abstract Book of participants of the International research and practice conference, 26–28 August 2026, Chernivtsi. Edited by Dr. Olena Fesenko. – Kyiv: 2026. – P. 662.

This book contains the abstracts of contributions presented at the International research and practice conference “Nanotechnology and Nanomaterials” (NANO-2026), held on 26–28 August 2026 at Yuriy Fedkovych Chernivtsi National University, Chernivtsi, Ukraine.

The NANO-2026 Conference was organized by the Institute of Physics of NAS of Ukraine with the participation of the University of Tartu (Estonia), Yuriy Fedkovych Chernivtsi National University, University of Turin (Italy) and Pierre and Marie Curie University – Paris 6 (France).

NANO-2026 is the 14th conference in the series of NANO-conferences initiated by the Institute of Physics of NAS of Ukraine in 2012 in the framework of FP7 Nanotwinning project. From year to year, they attract more attention and participants. In 2012, the first meeting was held in the format of International Summer School for young scientists «Nanotechnology: from fundamental research to innovations». The 2013 and 2014 conferences were organized in conjunction with the International Summer Schools for young scientists under the same title. In 2013, this event was attended by more than 300 scientists, in 2014-2017, 450 scientists took part and in 2018 it gathered above 650 participants. In 2021 conference was attended by more than 700 scientists from Ukraine, Poland, Italy, Estonia, France, Austria, Germany, Greece, Turkey, USA, Romania, Moldova, Czech Republic, Taiwan, Lithuania, Egypt, India, Algeria, Indonesia and other countries. In 2021 - 2025 the Organizer Committee has received more than 800 application forms from about 25 countries of the world each years.

The NANO-2026 conference brought together leading scientists and young researchers from many countries of the world. This year its topics were as follows: Nanobiotechnology for health-care; Nanochemistry and biotechnology; Nanocomposites and nanomaterials; Nanooptics and photonics; Nanoplasmonics and surface enhanced spectroscopy; Nanoscale physics; Nanostructured surfaces; Physico-chemical nanomaterials science.

Website of the NANO-2026 conference: <http://nano-conference.iop.kiev.ua>

Optimization of thermal transport properties in multilayer silicon structures via a hybrid photoacoustic and PINN approach

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Understanding heat transport in multilayer porous silicon (MPSi) structures is of great importance for modern micro- and nanoelectronics, thermoelectric systems, and advanced thermal insulation technologies, where precise control of thermal conductivity is essential to ensure device reliability. While single-layer porous silicon offers significant thermal-conductivity reduction, MPSi structures introduce complex phonon-scattering mechanisms at interfaces between layers of varying porosity. To address these challenges, this study reveals a comprehensive multiscale approach combining experimental measurements, molecular dynamics (MD) simulations, and machine learning.

The thermal transport properties of electrochemically etched MPSi structures with alternating porosity layers (ranging from 50% to 75%) were investigated using the gas-microphone photoacoustic (PA) method. The non-destructive PA technique provided the amplitude-frequency characteristics of signals, which were analyzed to determine the effective thermal conductivity of the structures.

To overcome the limitations of classical effective medium approximations, which fail to fully capture volumetric light absorption and morphological imperfections, a Physics-Informed Neural Network (PINN) was employed. The PINN models were trained by integrating the 2D Fourier heat conduction equation directly into the loss function, using realistic structural inputs generated via the Quartet Structure Generation Set algorithm, and were validated by Scanning Electron Microscopy. Furthermore, MD simulations utilizing the Green-Kubo method were performed to investigate the microscopic mechanisms of phonon transport. Both experimental and PINN-based data deviate from standard effective medium predictions, highlighting the critical role of interfacial thermal boundaries. This hybrid approach provides effective predictive models for the precise design and optimization of advanced thermal management components based on nanostructured silicon.

The work was supported by National Research Foundation of Ukraine (project No. 2023.03/0252).